A Framework for Spacetime from Weighted Histories

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October 12, 2026

Abstract

1 Introduction

2 What spacetime needs to be

Both theories of physics, GR and QFT, have their own requirements out of spacetime. The properties of spacetime needed by them are explored here.

2.1 General relativity

2.1.1 Manifold structure

Definition 2.1 (Topological manifold)**.** An n-dimensional topological manifold is a topological space M with three properties.

1. Every point of M has an open neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$.
2. M is Hausdorff.
3. M is second countable.

The first property says that if one zooms in close enough on any point, the space looks like ordinary $\mathbb{R}^n$, or a standard flat metric space. A homeomorphism is a continuous map with a continuous inverse, so each small patch can be labelled with n real numbers in a way where nearby points get nearby numbers. These labels are the coordinates. The surface of a sphere is the standard example. Every small piece of it looks like a flat sheet, but globally it is not a plane.

The Hausdorff property says two different points can always be put in separate open neighbourhoods. Without it one can build spaces that branch, like a line that splits into two copies of itself past some point.

Second countability says the topology can be generated by countably many open sets. It keeps the space from being absurdly large, and it implies paracompactness, which is what one needs to glue local constructions together. In particular it guarantees that a smooth manifold carries a Riemannian metric.

GR takes spacetime to be a manifold with $n = 4$. The four is put in by hand, since Einstein's equations work in any dimension of three or more. GR also takes the manifold to be connected, so it comes in one piece, and every point has a path connecting it to every other point.

2.1.2 Smoothness

Definition 2.2 (Chart and atlas)**.** Let M be an n-dimensional topological manifold. A chart on M is a pair (U, φ) where $U \subseteq M$ is open (the patch) and $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ is a homeomorphism onto an open set (the coordinates on that patch). Two charts (U, φ) and (V, ψ) are C^k-compatible if the transition map

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$$

and its inverse are k times continuously differentiable. A C^k atlas is a collection of pairwise C^k-compatible charts whose domains cover M.

Definition 2.3 (Smooth manifold)**.** A C^k structure on M is a maximal C^k atlas, meaning one that contains every chart compatible with all of its charts. A smooth manifold is a topological manifold together with a C^∞ structure. A function $f : M \to \mathbb{R}$ is smooth if $f \circ \varphi^{-1}$ is C^∞ for every chart (U, φ) in the structure.

Each patch of the manifold can be mapped to a piece of flat $\mathbb{R}^n$, but different patches get their own separate maps. Where two patches overlap, the same points get two sets of coordinates, and the transition map converts one set into the other. Asking it to be differentiable is what makes derivatives meaningful on M. If a function is differentiable in one coordinate system, it is differentiable in every other one, and the derivatives are related by the chain rule. Taking the maximal atlas means the smooth structure does not depend on which particular charts were written down first.

GR needs smoothness because gravity is curvature, and curvature is built from second derivatives of the metric. So the metric components have to be at least C^2 (twice continuously differentiable) in every chart, which in turn needs a manifold with at least a C^3 structure. In practice people assume both the manifold and the metric are C^∞. The singularities inside black holes and at the big bang are exactly the places where this assumption breaks down.

2.1.3 Metric and signature

Definition 2.4 (Metric)**.** Let M be a smooth manifold and let T_pM be the tangent space at $p \in M$ (the vector space of velocity vectors of smooth curves passing through p). A metric on M is a smooth assignment, to every point p, of a bilinear map $g_p : T_pM \times T_pM \to \mathbb{R}$ that is symmetric, $g_p(u, v) = g_p(v, u)$, and nondegenerate, so the only u with $g_p(u, v) = 0$ for all v is $u = 0$. The metric is Riemannian if $g_p(v, v) > 0$ for every nonzero v. It is Lorentzian if at every p there is a basis of T_pM in which g_p is the diagonal matrix $\mathrm{diag}(-1, 1, \ldots, 1)$.

A metric is a rule that gives the length of a small step in any direction at every point. The tangent space is the set of all directions one can step in from p, and $g_p(v, v)$ is the length squared of a step v. A Riemannian manifold is a smooth manifold together with a Riemannian metric. On a Riemannian manifold, like the sphere, every length squared is positive.

GR uses a Lorentzian metric $g_{\mu\nu}$ instead, with one minus sign and three plus signs, $(-1, 1, 1, 1)$ being the Minkowski metric. Now a small step can have positive, negative, or zero length squared, which makes it spacelike, timelike, or lightlike. The metric gives the proper time along a path. That one minus sign is the whole difference between time and space.

2.1.4 Light cones and a direction of time

Definition 2.5 (Causal character and light cone)**.** Let (M, g) be a Lorentzian manifold and $p \in M$. A nonzero vector $v \in T_pM$ is timelike if $g_p(v, v) < 0$, lightlike (or null) if $g_p(v, v) = 0$, and spacelike if $g_p(v, v) > 0$. It is causal if it is timelike or lightlike. The light cone at p is the set of lightlike vectors in T_pM. A curve in M is causal if its velocity vector is causal at every point along it.

Definition 2.6 (Time orientation)**.** A Lorentzian manifold (M, g) is time orientable if there is a continuous vector field T on M that is timelike everywhere. Such a T picks out the future. A causal vector v at p is future directed if $g_p(v, T_p) < 0$ and past directed otherwise.

Definition 2.7 (Causal order)**.** In a time oriented Lorentzian manifold, write $p \preceq q$ if $p = q$ or there is a future directed causal curve from p to q. The causal future of p is $J^+(p) = \{q : p \preceq q\}$, and the causal past $J^-(p)$ is defined the same way with the order reversed.

Light cones are a property of spacetime itself and comes straight out of the metric. At each point, the lightlike directions form a double cone in the tangent space, and the timelike directions are the ones inside it.

The physics enters as a rule about what moves through spacetime. Every massive particle travels along a timelike curve, and light travels along lightlike curves. That is the precise meaning of "nothing goes faster than light" in GR. The speed limit is not a separate assumption sitting next to the geometry. It is the statement that matter follows causal curves of the metric. So the light cones tell which events can influence which, and the relation $p \preceq q$ is exactly "p can influence q".

The metric alone does not say which half of each cone is the future. Time orientability is the extra global condition that one can make that choice continuously across all of spacetime. It is a property of the spacetime as a whole, and some Lorentzian manifolds do not have it. Without it the causal order $\preceq$ cannot even be defined, since "future directed" has no meaning.

2.1.5 Local flatness

Near any point one can choose coordinates in which the metric looks exactly like flat Minkowski space, at least to first order. In those coordinates the laws of physics look like special relativity.

This is the equivalence principle written in geometry. It means every tiny patch of spacetime has the Lorentz group as its symmetry, even when the spacetime as a whole has no symmetry at all. GR also needs a way to compare vectors at nearby points, called the connection. The connection gives geodesics, which are the paths that freely falling things follow.

2.1.6 Volume

A volume element is what one needs to integrate over a region of spacetime.

GR needs one because the Einstein–Hilbert action is an integral, $\int R\sqrt{-g}\, d^4x$. The metric supplies the volume element through $\sqrt{-g}$. The light cones on their own do not. If one multiplies the metric by a positive function that changes from point to point, every light cone stays the same but all the volumes change.

2.1.7 No fixed background

In GR the metric is not given in advance. It is whatever solves Einstein's equations, and it bends in response to the matter present. Coordinates are only labels. Moving the points of the manifold around while dragging the metric along with them changes nothing physical. This is called diffeomorphism invariance.

2.1.8 Global hyperbolicity

A Cauchy surface is a slice of spacetime that every timelike path crosses exactly once. A spacetime that has one is called globally hyperbolic.

Einstein's equations allow strange solutions, including ones with closed timelike curves that let one travel into one's own past. To predict anything one wants data on a slice to fix everything before and after it, and that needs a Cauchy surface. GR does not demand this. It is an extra condition people add when they want a well defined initial value problem.

2.1.9 Topology

Topology is the global shape of the manifold, the features that stay the same when one stretches it without tearing. Whether space closes up on itself or goes on forever, and whether it has holes or handles, are topological questions.

Einstein's equations are local, so they say almost nothing about topology. It has to be chosen separately.

2.2 Quantum field theory

2.2.1 Manifold structure

Ordinary QFT lives on Minkowski space, which as a manifold is just $\mathbb{R}^4$. QFT in curved spacetime allows other four-dimensional manifolds, but it almost always asks for them to be globally hyperbolic, for the reasons given below.

2.2.2 Smoothness

Fields appear in equations of motion with derivatives, so the manifold has to be smooth. Quantum fields are also too rough to have a value at a single point. They only make sense when smeared out against smooth test functions, which again needs a smooth manifold.

QFT also assumes the manifold stays smooth at arbitrarily short distances. That assumption is where the ultraviolet infinities come from. Renormalization is the machinery that handles them, and it needs a notion of length or energy scale to work with.

2.2.3 A fixed background metric

In QFT the metric is chosen before any physics happens. In ordinary QFT it is the flat Minkowski metric. In QFT in curved spacetime it can be curved, but it is still fixed in advance. The fields live on this spacetime and do not bend it.

2.2.4 Poincaré symmetry

The Poincaré group is the set of translations, rotations, and boosts of Minkowski space. These are the transformations that leave the flat metric unchanged.

Ordinary QFT requires its laws to be invariant under this group. Translation symmetry gives conservation of energy and momentum. The bigger payoff is that particles are defined by this symmetry. A particle species is an irreducible unitary representation of the Poincaré group, and these are labelled by mass and spin. The spin–statistics theorem and CPT invariance also depend on Lorentz symmetry.

2.2.5 Causal structure

Measurements of fields at spacelike separated points must not affect each other. In the math, field operators at spacelike separation commute. So QFT needs to know, for every pair of points, whether they are spacelike separated. This is exactly the light cone information that the metric provides.

2.2.6 A time direction and positive energy

To write a Hamiltonian, QFT needs a notion of time translation. It then demands that energy is bounded below, so there is a lowest energy state, the vacuum, and the world is stable.

In flat space the vacuum is unique and every inertial observer agrees on it. In a curved spacetime with no time symmetry there is no preferred time, so there is no preferred vacuum.

Different observers can then disagree about how many particles are present. This is what is behind the Unruh and Hawking effects.

2.2.7 Regions and locality

QFT attaches observables to regions of spacetime. A measurement done in some region is described by operators belonging to that region, and bigger regions contain the observables of smaller ones.

Knowing the fields in a thin slab around a Cauchy surface determines them everywhere. That is why QFT in curved spacetime asks for global hyperbolicity.

2.2.8 Spin structure

Electrons and quarks are described by spinor fields. A spinor flips sign under a full 360° rotation and only comes back to itself after 720°.

Defining spinors consistently over the whole manifold is only possible when its topology allows it. This requirement is called a spin structure. A spacetime without one cannot host fermions.

2.2.9 Dimension

QFT can be written in any dimension. Which interactions give finite, sensible predictions after renormalization does depend on the dimension, though. In four dimensions the gauge theories of the Standard Model work. In higher dimensions they generally do not.

3 Search for the perfect space

4 Results

5 Discussion

References